MODULE 2 SKULL

The Skull.

AXIAL SKELETON

The skull is 22 bones in two groups, the cranium that encloses the brain and the facial skeleton that frames the face. This chapter names every bone, the sutures and fontanelles, the key markings, the cavities, and the major foramina.

BY THE END

1. Distinguish the cranial bones from the facial bones and name the bones in each group.
2. Identify the significant markings on each cranial and facial bone, and the major sutures and fontanelles.
3. Identify the bones that form the orbit, the nasal cavity, and the hard palate, and locate the paranasal sinuses.
4. Name the major foramina of the skull and the structure each one transmits.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Axial Skeleton, Skull

drsrennie-stack.github.io/new-build-bio4-solano/axial-concept-videos.html

The Skull, an Overview

The skull has 22 bones. The eight cranial bones enclose the brain, and the fourteen facial bones frame the face. The hyoid bone of the neck is associated with the skull but is counted separately.

ORIENTATION TERMS

Term	What it is
The skull	The bony framework of the head, 22 bones, plus the associated hyoid bone
Cranium	The eight cranial bones that enclose and protect the brain
Facial skeleton	The fourteen facial bones that frame the face and hold the teeth
Calvaria	The domed roof of the cranium, the skullcap
Cranial base	The floor of the cranium, shaped into anterior, middle, and posterior cranial fossae

The Cranial Bones

Eight bones form the neurocranium, the braincase. The frontal, occipital, sphenoid, and ethmoid are single. The parietal and temporal bones are paired.

THE EIGHT AT A GLANCE

Bone	Number	Region it forms
Frontal bone	Single	The forehead, the roof of the orbits, and the floor of the anterior cranial fossa
Parietal bones	Paired	The superior and lateral walls of the cranium
Temporal bones	Paired	The inferior lateral walls and part of the cranial base. House the ear
Occipital bone	Single	The posterior wall and much of the cranial base
Sphenoid bone	Single	The central wedge of the cranial base. Articulates with every other cranial bone
Ethmoid bone	Single	The deepest cranial bone, between the orbits. Forms part of the nasal cavity

Frontal bone

MARKINGS

Marking	What it is
Squamous part	The broad plate of the forehead
Supraorbital margin	The ridge above each orbit
Supraorbital foramen	The opening in that margin. Passes the supraorbital nerve and vessels
Glabella	The smooth area between the eyebrow ridges
Orbital plates	The paired plates that roof the orbits and floor the anterior cranial fossa
Frontal sinus	The air space within the bone, above the orbits

Parietal bones

MARKINGS

Marking	What it is
The cranial roof	The pair form most of the superior and lateral walls of the cranium
Parietal eminence	The rounded high point of each bone, the skull's widest measure
Suture borders	They meet at the sagittal suture and form the coronal, lambdoid, and squamous sutures with neighbouring bones

Temporal bones

MARKINGS

Marking	What it is
Squamous part	The flat region forming the lower lateral wall of the skull
Zygomatic process	The bar that joins the zygomatic bone to complete the zygomatic arch
Mandibular fossa	The socket that receives the mandible, forming the temporomandibular joint

Marking	What it is
External acoustic meatus	The opening of the ear canal
Mastoid process	The bump behind the ear, a muscle attachment, filled with air cells
Styloid process	The slender spike below the ear that anchors tongue and neck muscles
Petrous part	The dense wedge of the cranial base that houses the middle and inner ear
Internal acoustic meatus	The canal carrying the nerves for hearing and facial movement
Stylomastoid foramen	The exit of the facial nerve, between the styloid and mastoid processes

Occipital bone

MARKINGS

Marking	What it is
Foramen magnum	The large central opening for the spinal cord
Occipital condyles	The paired knobs beside the foramen magnum that articulate with the atlas
External occipital protuberance	The midline bump felt at the back of the head
Hypoglossal canal	The canal passing the hypoglossal nerve, cranial nerve XII, to the tongue muscles
Foramen magnum contents	The medulla oblongata, the vertebral arteries, the anterior and posterior spinal arteries, the spinal accessory nerve (CN XI), and the alar and apical ligaments of the dens
Atlanto-occipital joint	The occipital condyles meeting the superior articular facets of the atlas, C1. The nodding joint that says yes

MNEMONIC

For the contents of the foramen magnum: VAMPires Sing AT Midnight. Vertebral arteries, Anterior and posterior spinal arteries, Membranes and ligaments, Spinal accessory nerve, Medulla.

Sphenoid bone

MARKINGS

Marking	What it is
Body	The central cube of the bone. Contains the sphenoid sinus
Sella turcica	The saddle on the body whose deep seat cradles the pituitary gland
Greater wings	The broad lateral wings forming part of the cranial floor, the orbit, and the lateral skull wall
Lesser wings	The smaller anterior wings forming part of the anterior cranial fossa
Pterygoid processes	The downward projections that anchor the chewing muscles
Optic canal	Passes the optic nerve from the eye
Superior orbital fissure	Passes the nerves that move the eye

Ethmoid bone

MARKINGS

Marking	What it is
Crista galli	The upright ridge where the falx cerebri, a fold of the brain covering, attaches
Cribriform plate	The perforated horizontal plate that the olfactory nerve filaments pass through
Perpendicular plate	The vertical plate forming the upper part of the nasal septum
Lateral masses	The side portions, which contain the ethmoidal air cells
Superior and middle conchae	The scroll-shaped projections into the nasal cavity

The Facial Bones

Fourteen bones form the viscerocranium, the facial skeleton. Six are paired, and two, the vomer and the mandible, are single.

THE FOURTEEN AT A GLANCE

Bone	Number	What it forms
Maxillae	Paired	The upper jaw. Form the anterior hard palate and hold the upper teeth
Palatine bones	Paired	The posterior hard palate and part of the orbit
Zygomatic bones	Paired	The cheekbones
Nasal bones	Paired	The bridge of the nose
Lacrimal bones	Paired	The medial wall of the orbit, grooved for the tear duct
Inferior nasal conchae	Paired	Scroll-shaped bones on the lateral wall of the nasal cavity
Vomer	Single	The inferior part of the nasal septum
Mandible	Single	The lower jaw, the only freely movable bone of the skull

The maxillae

MARKINGS

Marking	What it is
Body	The central mass of the bone. Contains the maxillary sinus
Alveolar process	The ridge that holds the upper teeth
Palatine process	The shelf forming the anterior part of the hard palate
Frontal process	The upward projection that meets the frontal bone
Zygomatic process	The lateral projection that meets the cheekbone
Infraorbital foramen	The opening below the orbit. Passes the infraorbital nerve to the face

The mandible

MARKINGS

Marking	What it is
Body	The horizontal, U-shaped portion that holds the lower teeth
Ramus	The vertical portion on each side
Condylar process	The posterior process whose head meets the temporal bone at the jaw joint
Coronoid process	The anterior process where the temporalis chewing muscle attaches
Mandibular notch	The gap between the condylar and coronoid processes
Alveolar process	The ridge that holds the lower teeth
Mandibular foramen	The opening on the inner ramus. The entry for the nerve to the lower teeth, and the target for a dental block

Sutures and Fontanelles

The major sutures

A suture is an immovable fibrous joint between skull bones.

BONES JOINED AND LOCATION

Suture	Bones it joins	Location
Coronal suture	The frontal bone and the two parietal bones	Runs across the top of the skull, behind the forehead
Sagittal suture	The two parietal bones	The midline of the skull roof
Lambdoid suture	The parietal bones and the occipital bone	The back of the skull
Squamous suture	A parietal bone and a temporal bone	The lower side of the skull, one on each side
Pterion	The frontal, parietal, temporal, and sphenoid bones	The H-shaped junction on the side of the skull. The thinnest part

The fontanelles

Fontanelles are fibrous membrane gaps between the bones of an infant skull. They let the skull compress during birth and grow with the brain.

LOCATION AND CLOSURE

Fontanelle	Also called	Location	Closes by
Anterior fontanelle	The largest fontanelle	Where the frontal and parietal bones meet	About 18 to 24 months
Posterior fontanelle	The occipital fontanelle	Where the parietal bones meet the occipital bone	About 2 months
Anterolateral fontanelles	The sphenoidal fontanelles	On the side of the skull, one on each side	About 3 months
Posterolateral fontanelles	The mastoid fontanelles	Behind the ear, one on each side	Begin at 1 to 2 months, complete by about 12 months

Cavities and Special Features

The orbit and nasal cavity

WHAT BUILDS EACH SPACE

Structure	What it is
The orbit	The bony eye socket, built from seven bones: frontal, sphenoid, zygomatic, maxilla, lacrimal, ethmoid, and palatine
Nasal cavity	The space behind the nose, with a bony roof, floor, lateral walls, and septum
Nasal septum	The partition of the nasal cavity. The vomer plus the perpendicular plate of the ethmoid plus septal cartilage
Hard palate	The bony roof of the mouth. The maxillae in front and the palatine bones behind

Paranasal sinuses

Air-filled cavities that lighten the skull and drain into the nasal cavity.

FOUR SINUSES, BY THE BONE EACH OCCUPIES

Sinus	Bone it occupies	Location
Frontal sinus	The frontal bone	Above the orbits
Ethmoidal sinuses	The ethmoid bone	The ethmoid air cells, between the orbits
Sphenoidal sinus	The sphenoid bone	Within the body of the sphenoid
Maxillary sinuses	The maxillae	The largest sinuses, in the cheek region

The hyoid bone

THE BONE THAT TOUCHES NO OTHER BONE

Feature	What it is
Hyoid bone	A U-shaped bone in the anterior neck, below the mandible
Role	Anchors the tongue and several muscles of the neck and larynx
Unique feature	It does not articulate with any other bone. It is slung by ligaments from the styloid processes

Major Foramina

A foramen is an opening that lets a nerve or a vessel pass through bone.

BONE PIERCED AND WHAT PASSES THROUGH

Foramen	Bone	What passes through
Foramen magnum	Occipital bone	The spinal cord and the vertebral arteries
Optic canal	Sphenoid bone	The optic nerve, cranial nerve II
Superior orbital fissure	Sphenoid bone	Nerves that supply the eye, including those that move it
Foramen ovale	Sphenoid bone	A branch of the trigeminal nerve to the lower jaw
Cribriform foramina	Ethmoid bone	The olfactory nerve filaments, cranial nerve I
Internal acoustic meatus	Temporal bone	The nerves for hearing and for facial movement
Carotid canal	Temporal bone	The internal carotid artery
Jugular foramen	Between the temporal and occipital bones	The internal jugular vein
Mental foramen	Mandible	The nerve and vessels to the chin and lower lip

WHERE THE SKULL IS THIN, AND WHERE IT HAS ONLY ONE DOOR

The pterion, the H-shaped junction of the frontal, parietal, temporal, and sphenoid bones, is the thinnest part of the skull, and the middle meningeal artery runs just beneath it. A blow there can tear the artery and bleed into the skull. The paranasal sinuses drain into the nasal cavity through narrow openings, so a head cold can spread into them as sinusitis, and the maxillary sinus sits so close to the upper teeth that a tooth infection can reach it. A newborn is checked at the anterior fontanelle: a sunken fontanelle can signal dehydration, a bulging one can signal raised pressure inside the skull. The foramen magnum is the one large opening between the brain and the spinal cord, so when pressure builds inside the rigid skull, brain tissue can be forced down through it. That is why rising intracranial pressure is an emergency.